

Bubble sort

Revise 360

Worksheet for the 360° experience: Bubble sort conveyor

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Bubble sort conveyor. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Describe adjacent comparisons and swaps.
- Trace complete bubble-sort passes.
- Explain an early stopping condition.

Starter: before you put the headset on

For ascending order, should neighbouring values 7 and 3 be swapped? Why?

.....

.....

Key terminology

Pass

.....

.....

Swap

.....

.....

The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Compare neighbours

Turn right

Key facts in your own words:

Example to trace:

Ascending order

Compare: 7, 3

Swap: 3, 7

Compare: 3, 5 -> no swap

Challenge: Should equal neighbouring values be swapped?

Which pair needs swapping for ascending order?

Comparing adjacent values means comparing...

2 Complete the first pass

Turn right

Key facts in your own words:

Example to trace:

Start: 5, 1, 4, 2

After pair 1: 1, 5, 4, 2

After pair 2: 1, 4, 5, 2

After pair 3: 1, 4, 2, 5

Challenge: Is the list fully sorted after this pass?

Order the states in the first left-to-right pass.

Put these steps in order: 1, 4, 2, 5 / 1, 4, 5, 2 / 1, 5, 4, 2 / 5, 1, 4, 2

After the first pass on [5,1,4,2], the final item is...

3 Continue the passes

Turn around

Key facts in your own words:

Example to trace:

After pass 1: 1, 4, 2, 5

Pass 2: compare 1 and 4

Then swap 4 and 2

After pass 2: 1, 2, 4, 5

Challenge: What is the list after the second pass?

After pass 1 gives [2, 3, 1, 4], what does pass 2 give?

After two passes on [5,1,4,2], the list is...

4 Swaps and comparisons

Turn around

Key facts in your own words:

Example to trace:

Start: 1, 2, 4, 3
Compare 1/2: no swap
Compare 2/4: no swap
Compare 4/3: swap

Challenge: How many comparisons and swaps occur?

First full pass on [1, 2, 3, 4]: how many swaps?

A comparison always produces a swap. Is this true?

5 Read the swap

Turn left

Key facts in your own words:

Example to trace:

```
temp = a[i]
a[i] = a[i+1]
a[i+1] = temp
```

Challenge: Why is temp needed in this three-line swap?

Order the three-line swap.

Put these steps in order: Copy temp into a[i+1] / Copy a[i+1] into a[i] / Save a[i] in temp

Without saving a value before overwriting it, a swap may...

6 Stop when unchanged

Turn left

Key facts in your own words:

Example to trace:

Start pass: swapped = false
On a swap: swapped = true
No swaps in whole pass?
Stop: list is sorted

Challenge: Why is one correct pair not enough to stop?

Which is a safe early stopping condition?

When should swapped be reset to false?

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

First full left-to-right bubble pass on [4, 3, 2, 1] gives...

Why can the rightmost value then be excluded?

Your score

Copy it from the score screen as soon as it appears.

_____ / 19

Key question

Can moving one item at a time eventually put a whole list in order?

Review and improve

Use review mode. Explain one correction and why the correct answer fits.

Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Show the first left-to-right pass on [6, 2, 5, 1], recording each swap. [3]

b) Explain how a swap flag can reduce unnecessary passes. [2]

c) Explain why one pass does not necessarily sort a list. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Describe adjacent comparisons and swaps.			
Trace complete bubble-sort passes.			
Explain an early stopping condition.			